

Mutations

Evolution & Ecology

Jacopo Grilli, Ciudad de Guatemala, 26/11/2025

Plan

~~L1 - Exponential growth~~

L2 - Mutations

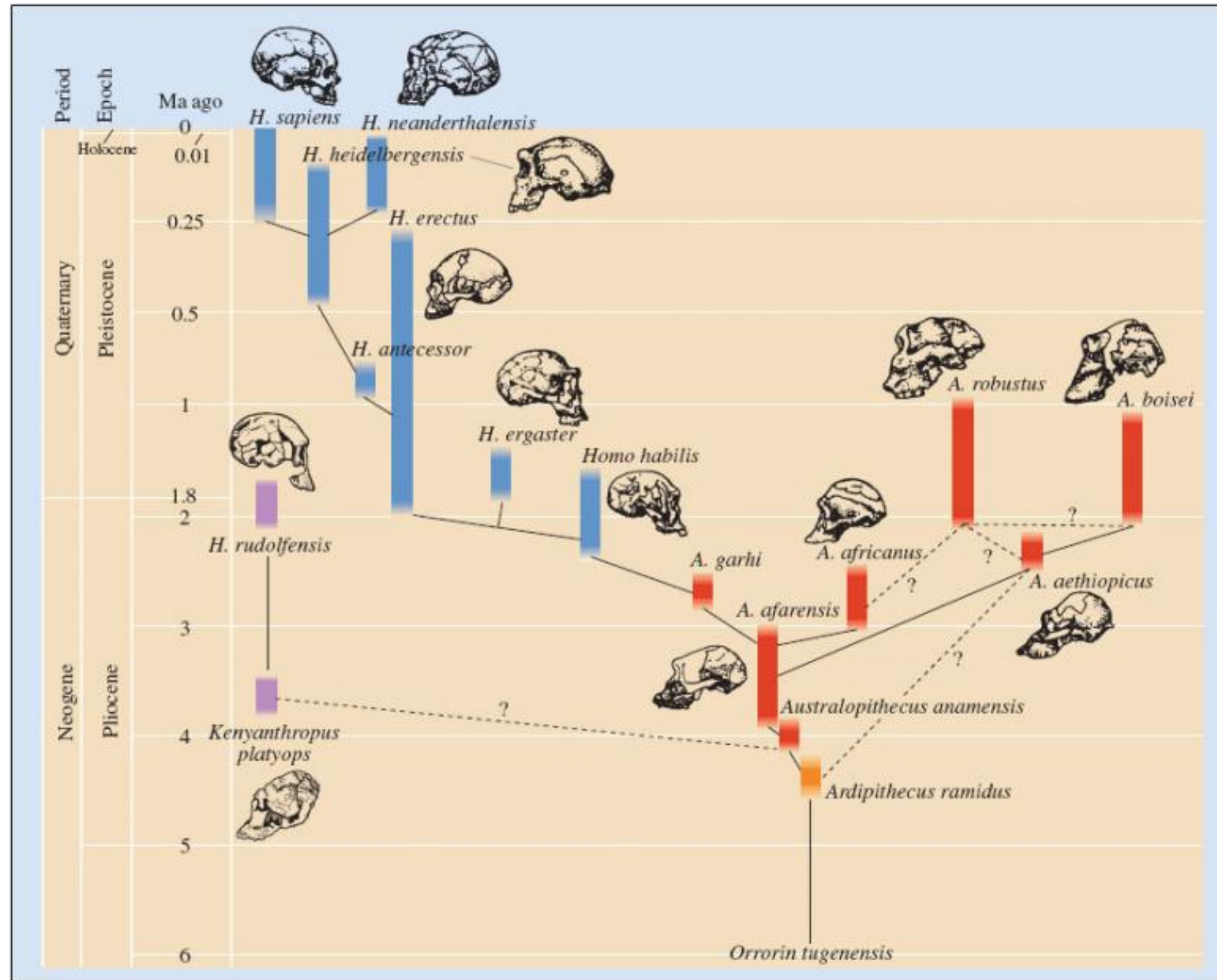
L3 - Selection

L4 - Ecology

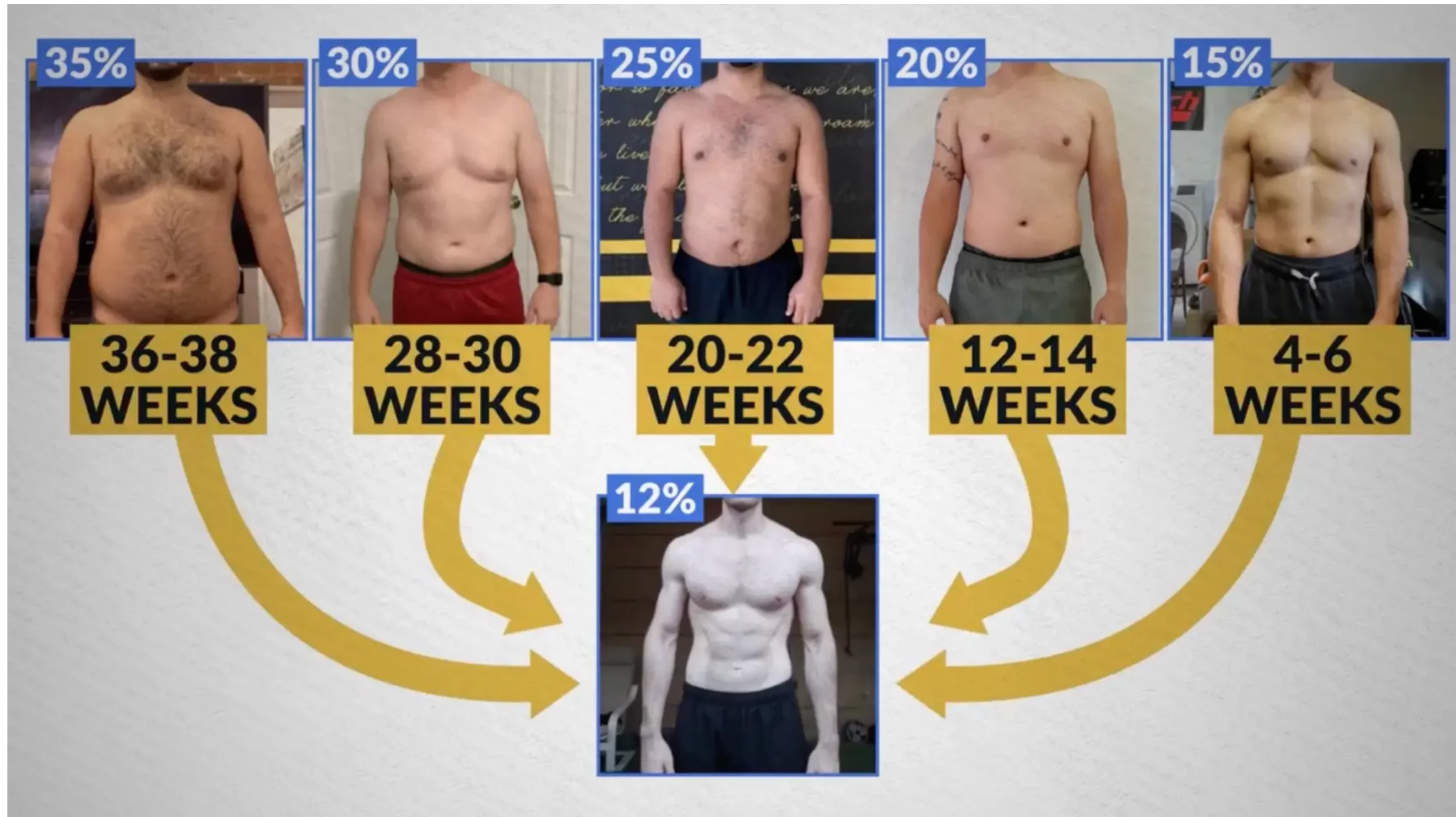
L5 - Eco-Evo

What is Evolution?

Evolution is about variation and change

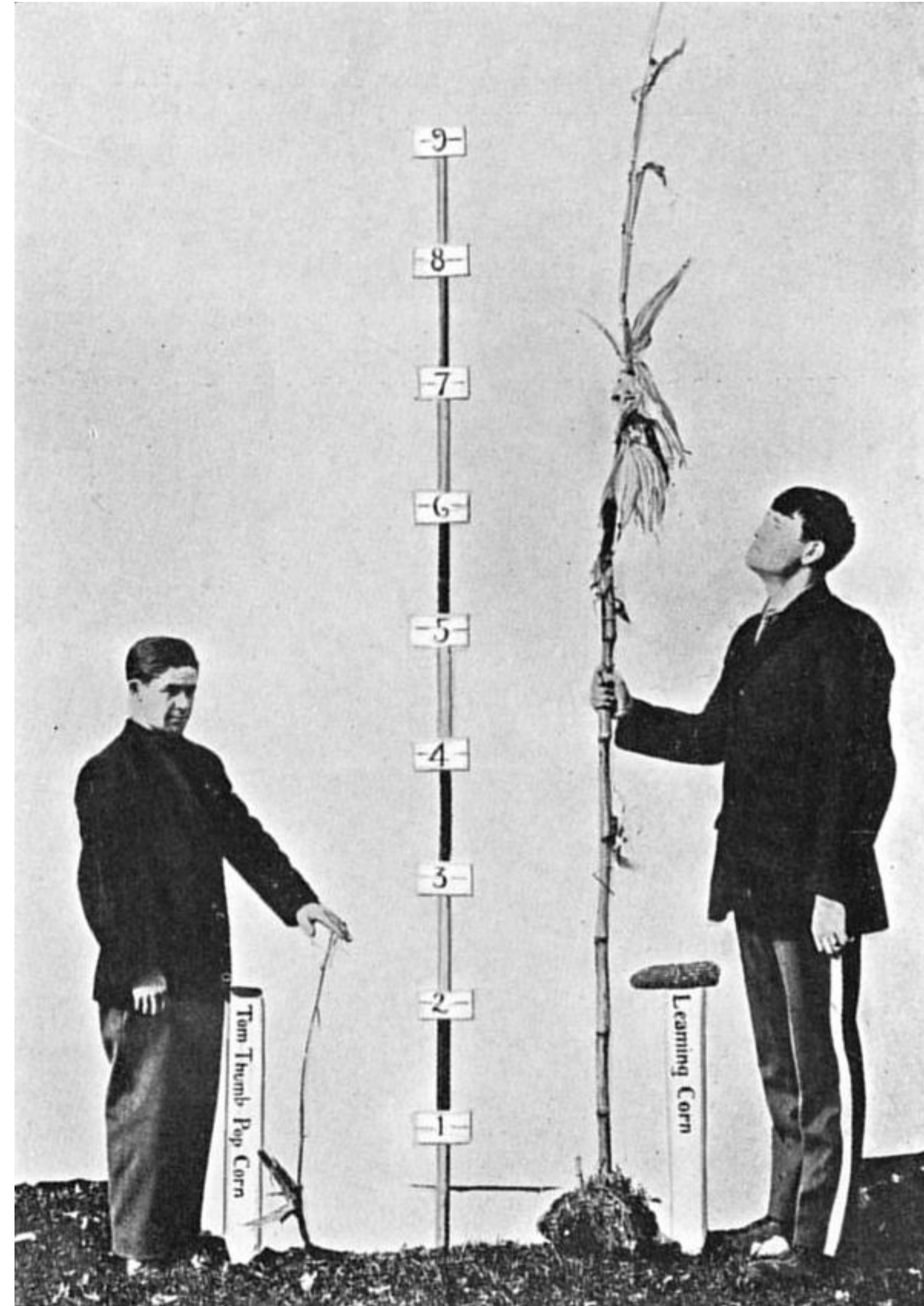
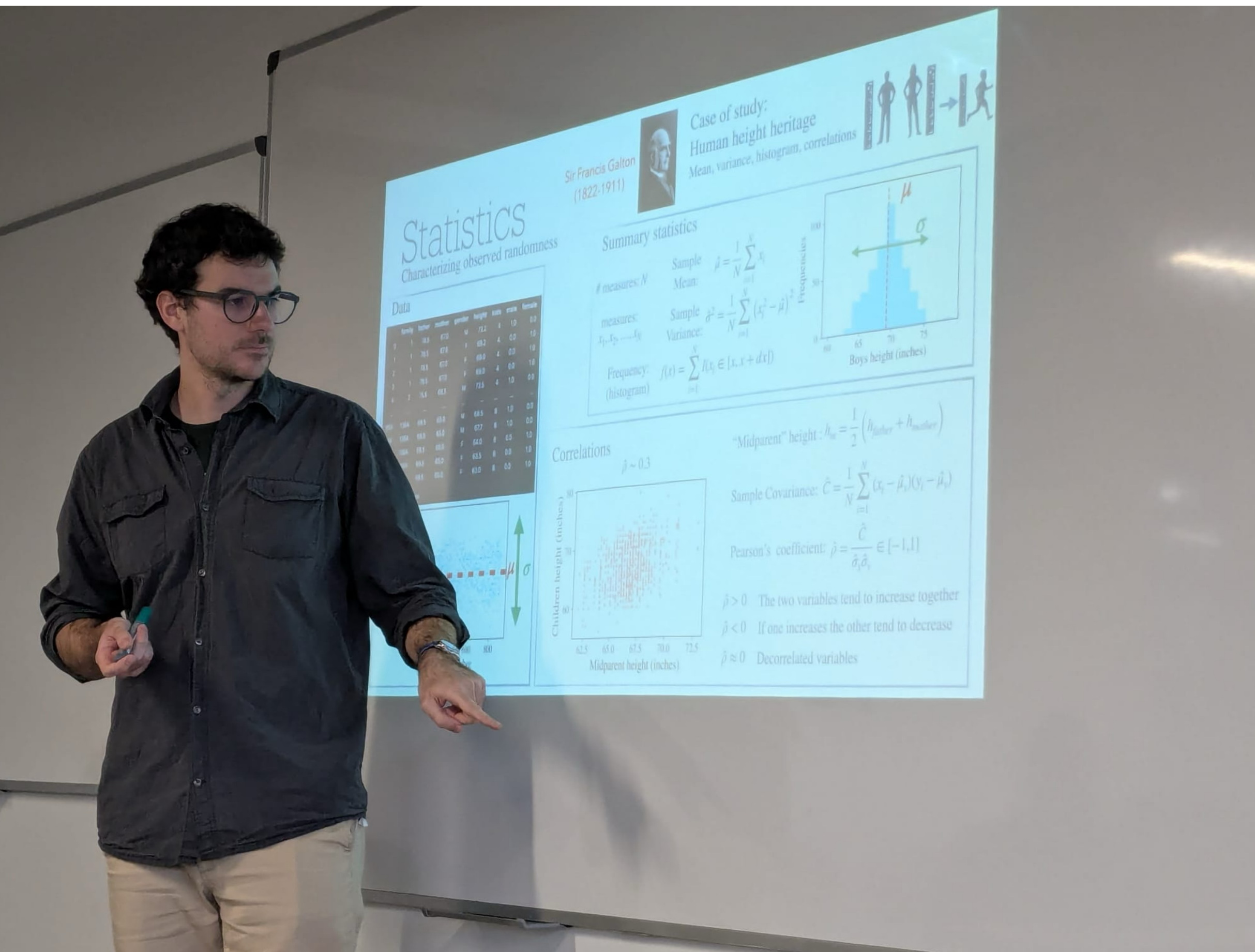


What type of change?

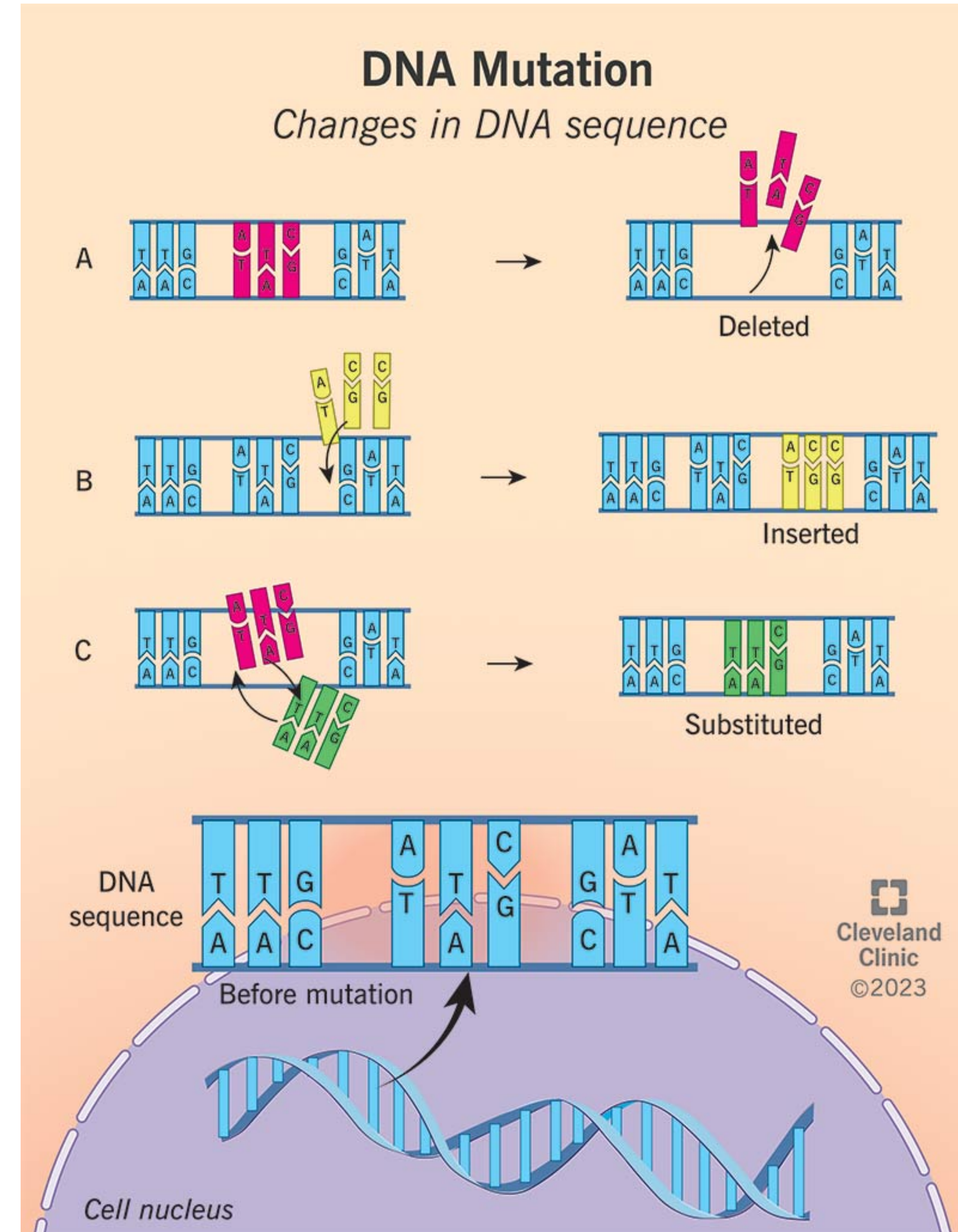


Evolution is about heritable variation

heritable = passed from parent(s) to offsprings



Mutations is one source of heritable variation



Selection acts on the existing variation

some heritable traits make some individuals more likely to reproduce than others

One outcome of selection is adaptation



One outcome of selection is adaptation



Evolution in action

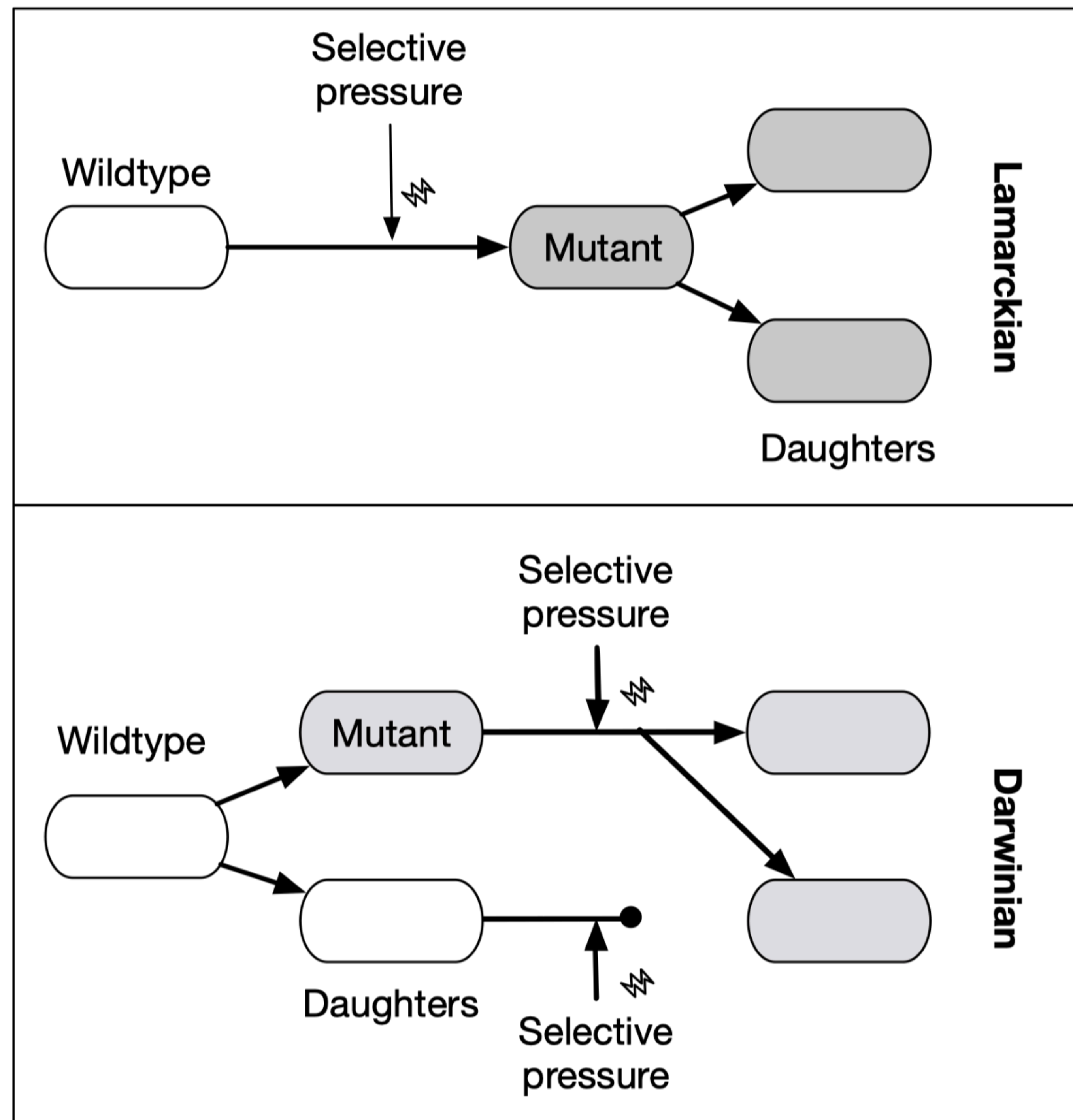


How do we know that this is about mutations?



It could be a response to a challenge

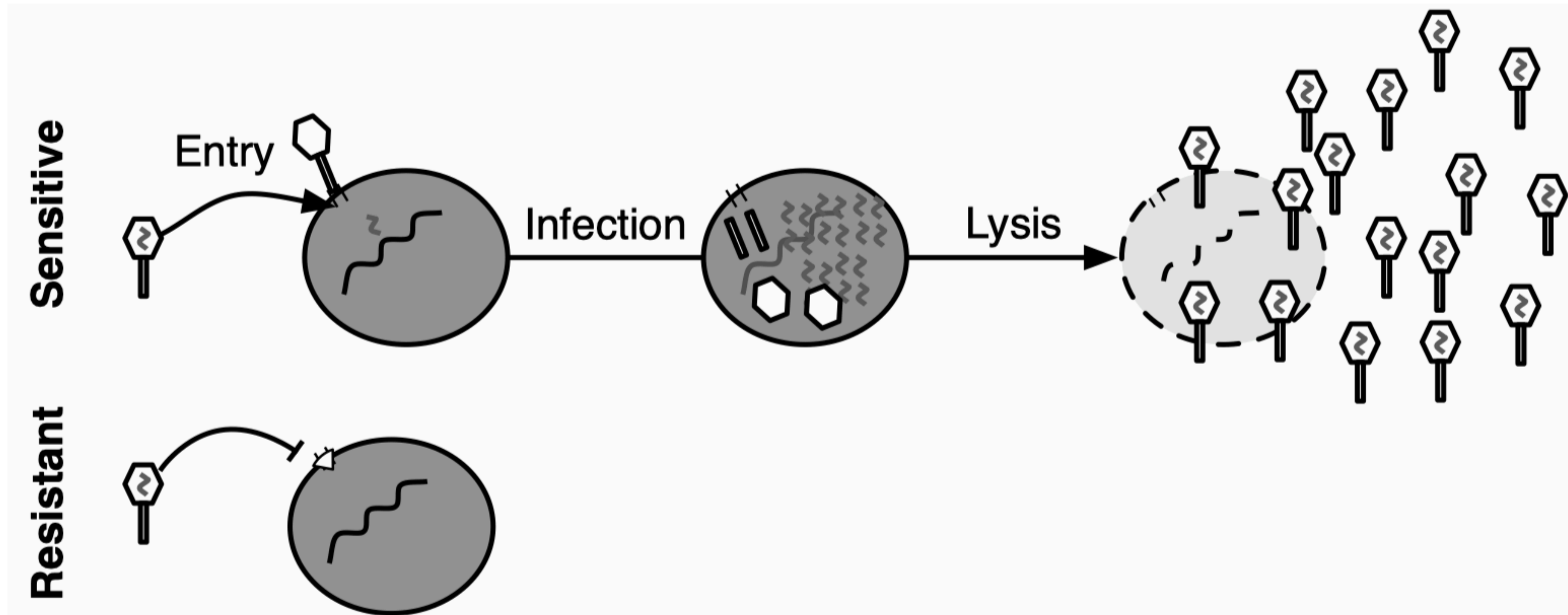
Two views of evolution



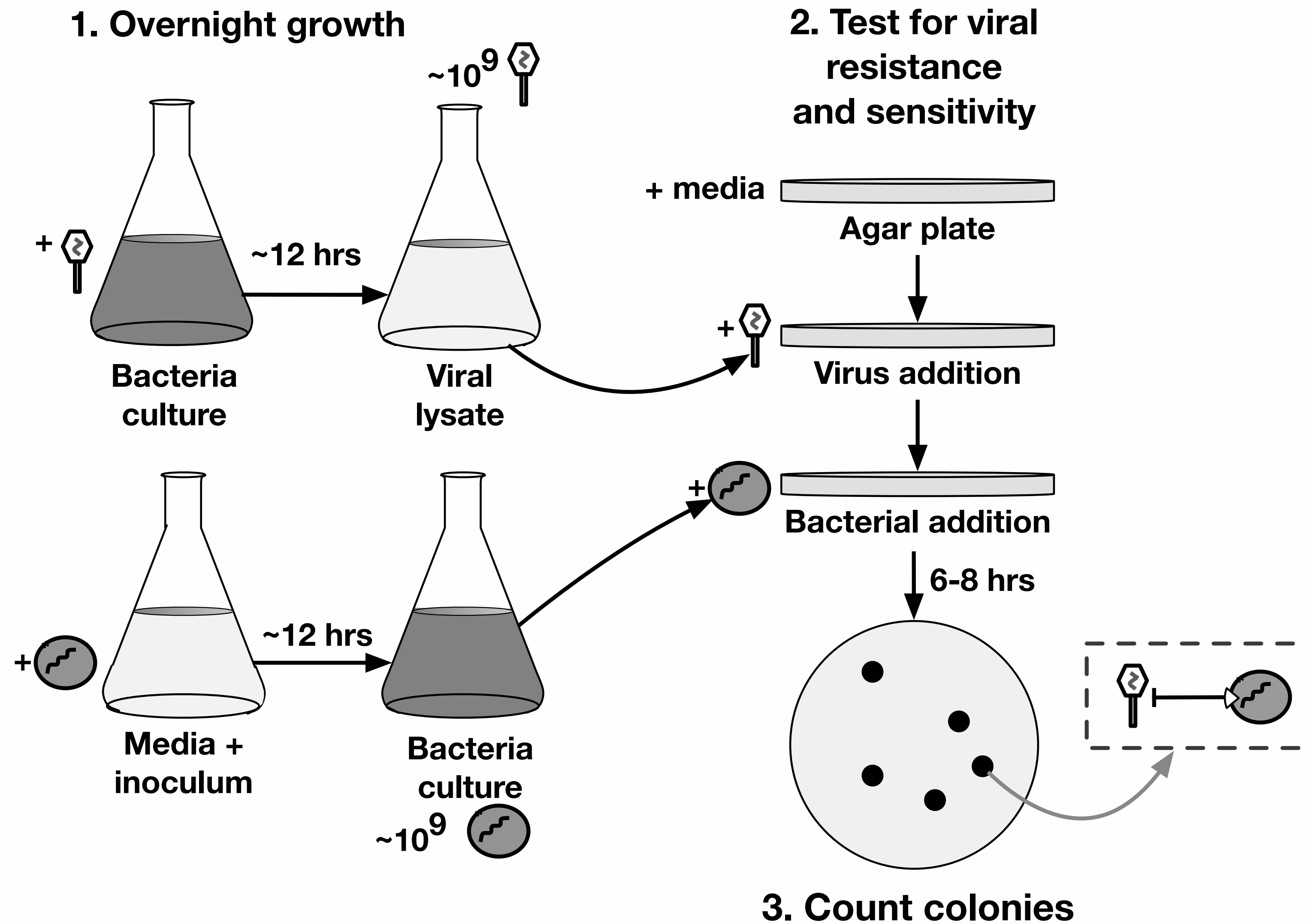
Luria and Delbrück



One slide about bacteria and phages



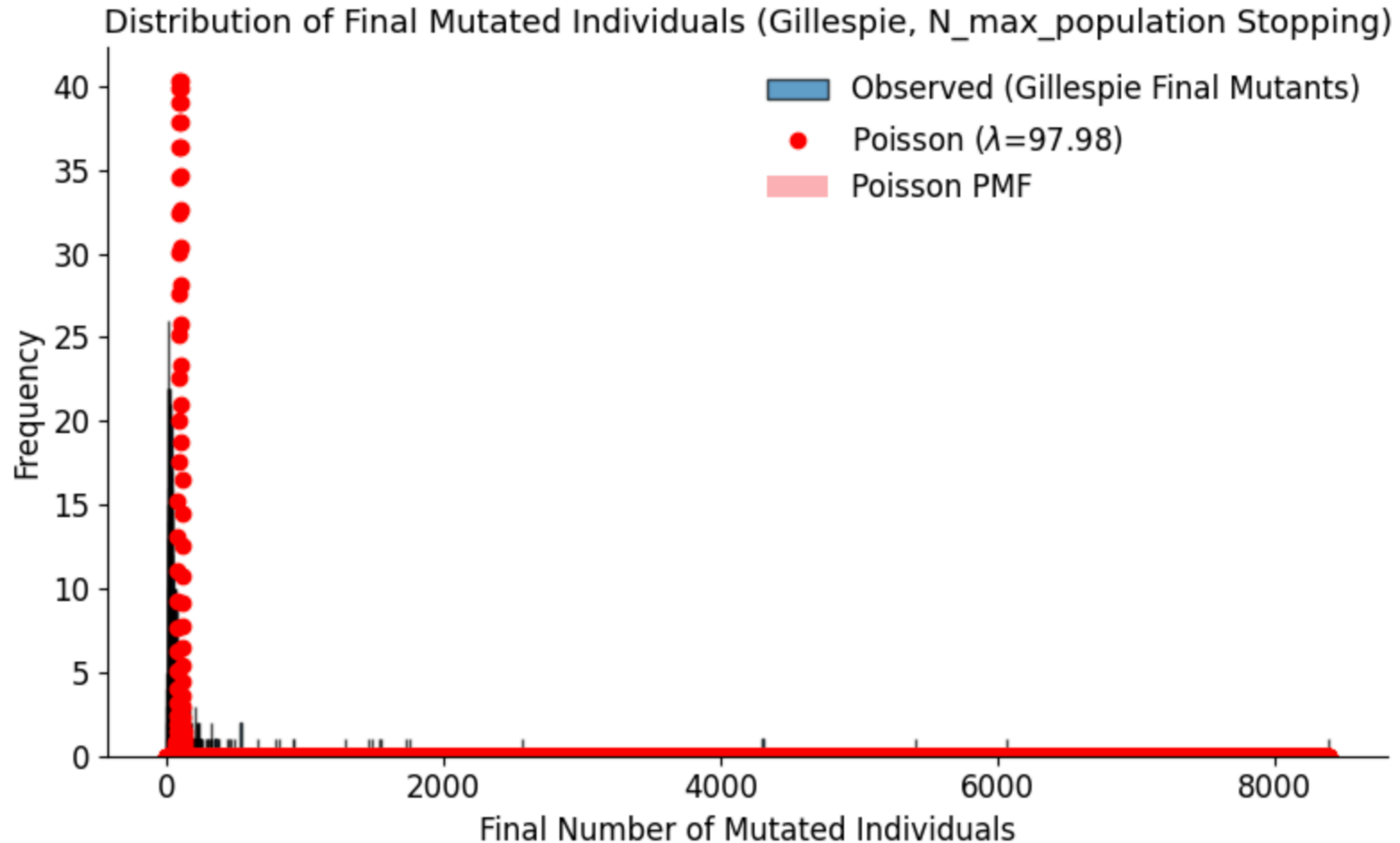
Luria-Delbrück experiment



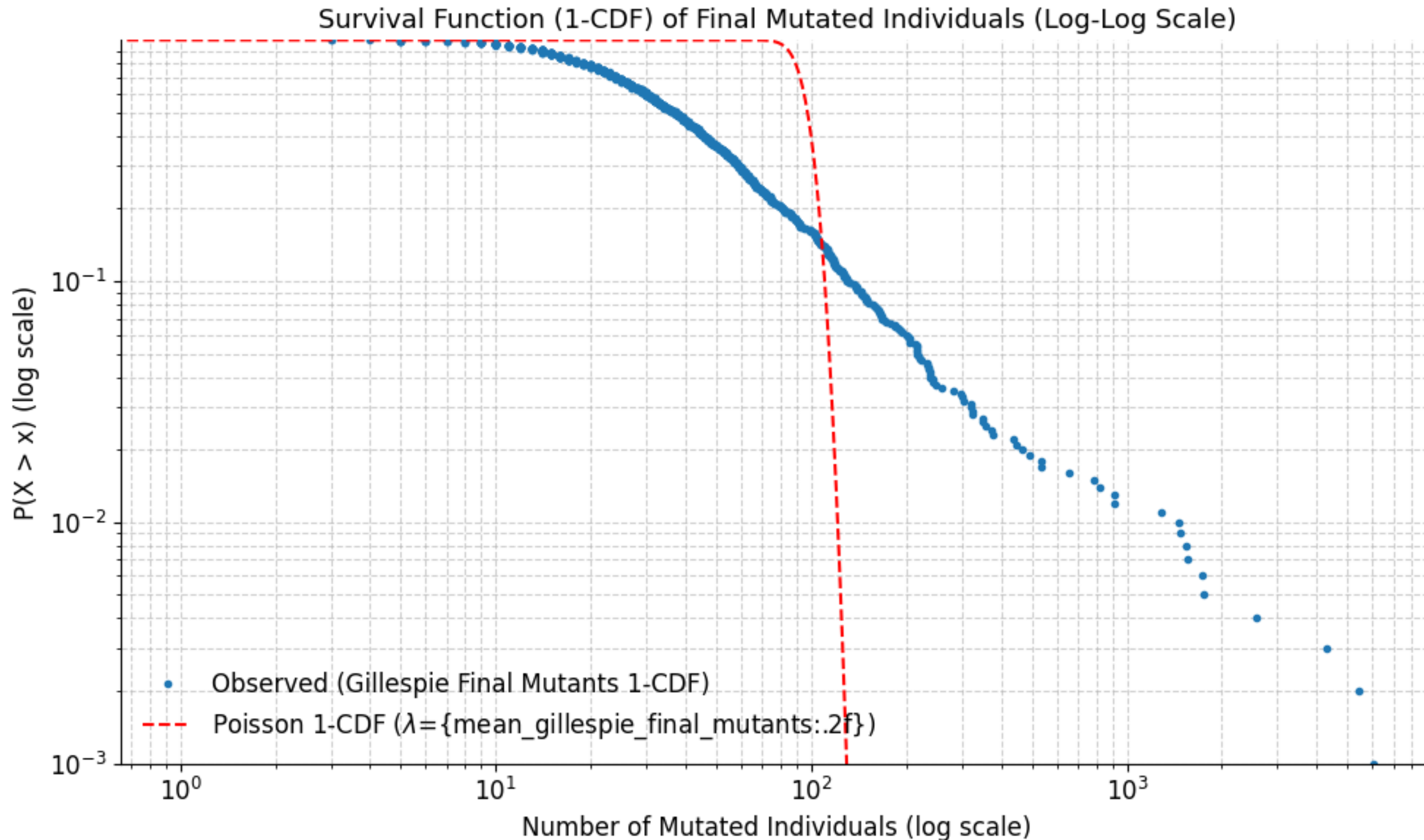
How many colonies should you expect?

let's count!

Simulation Results



Simulation Results



Initial Culture

First Generation

Second Generation

Third Generation

Fourth Generation

Culture 1



Spontaneous
Mutation



Second
Spontaneous
Mutation



Culture 2



Culture 3

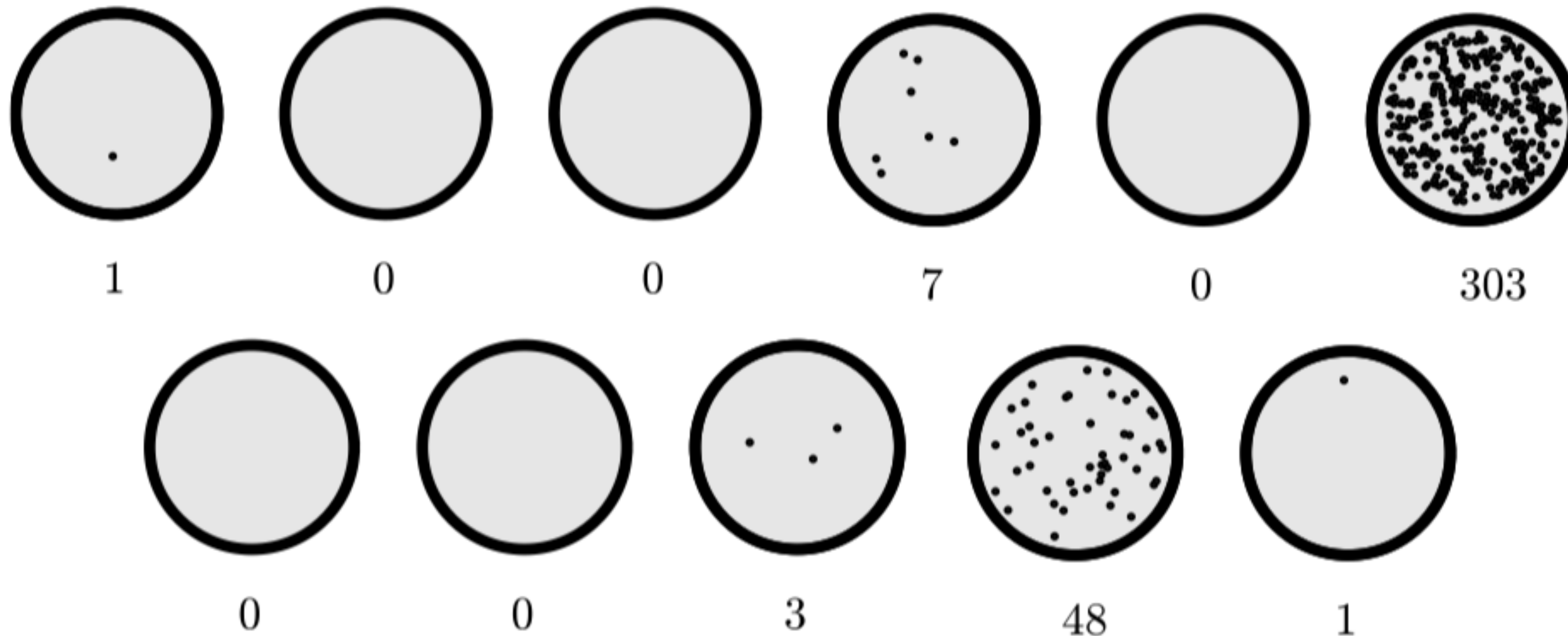


Spontaneous
Mutation



~~How many colonies should you expect?~~

What is the variability of the number of colonies?



The original experimental data

[illegible]

A modern view

A

clone

Ancestor Sequencing

GAT AAT CT GG TCT T A T T C C

~ 6 months
mutation-accumulation line
experiment
(see Figure 4)

clone

~ 6 weeks

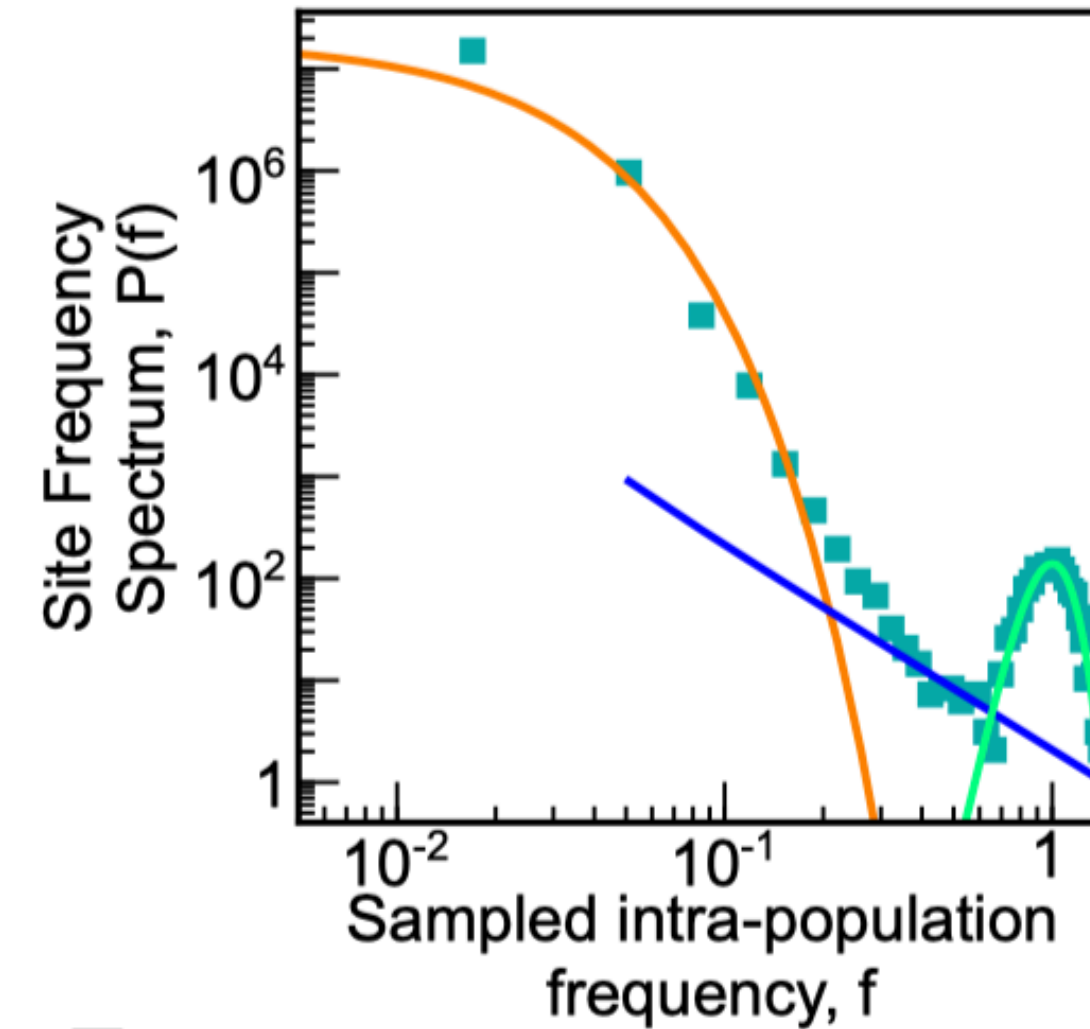
Endpoint Sequencing

GAT AAT CT GG TCT T A T T C C

subclonal
mutations

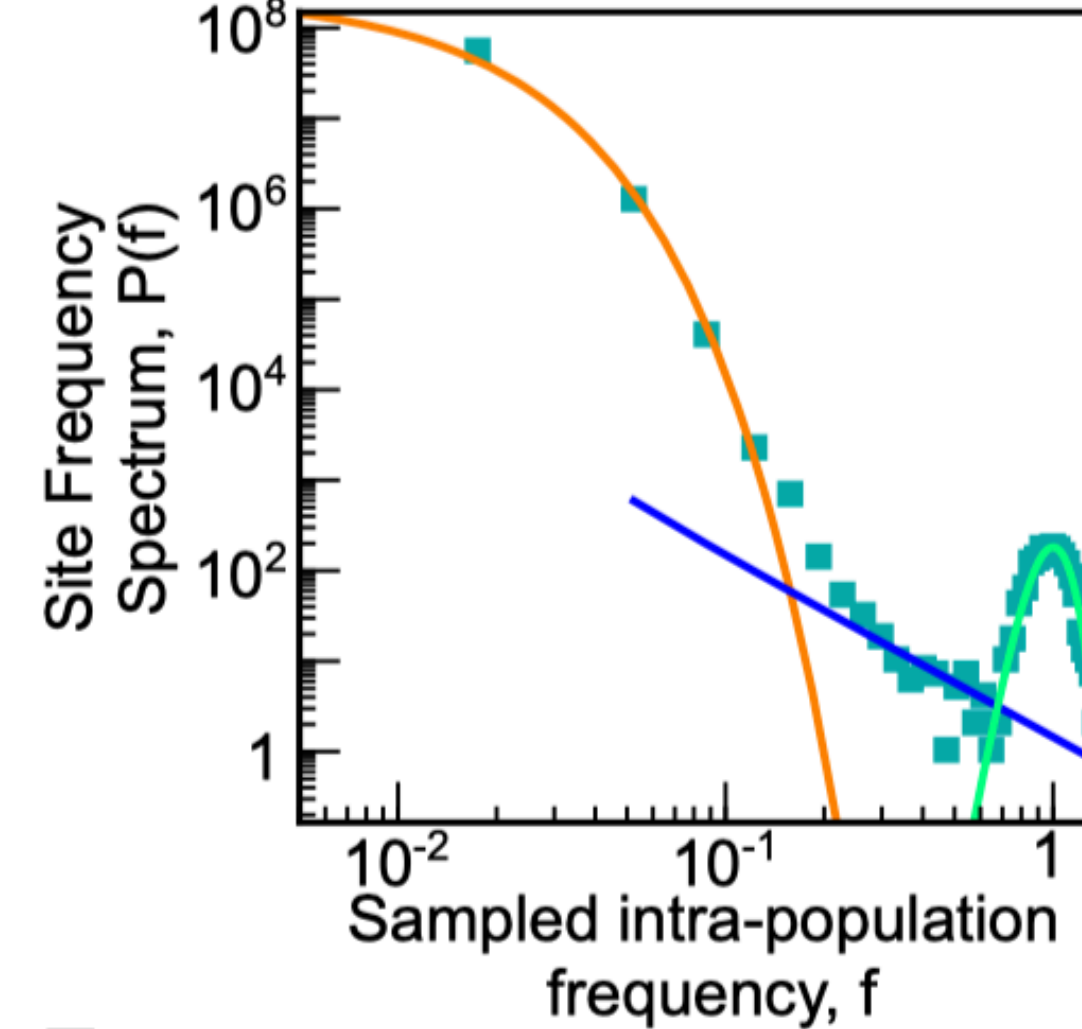
B

CRC1307LM (MSS)



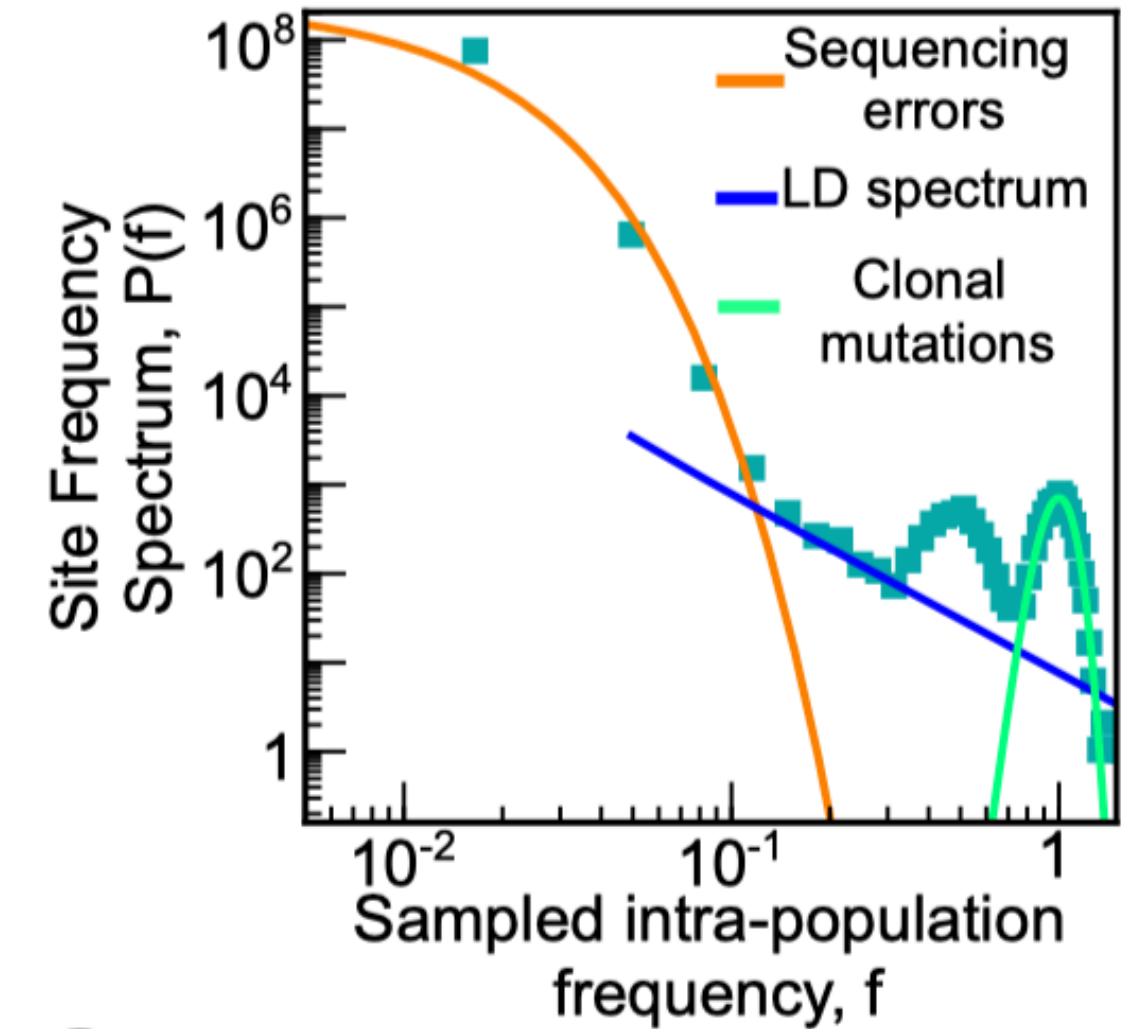
C

CRC1502LM (MSS)

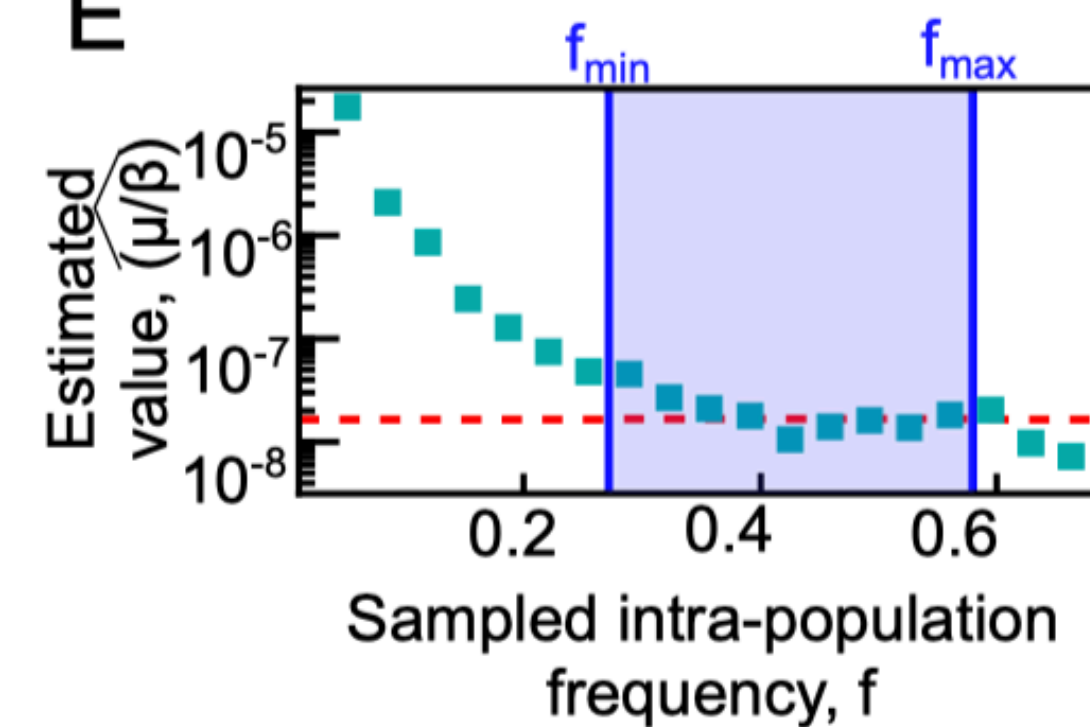


D

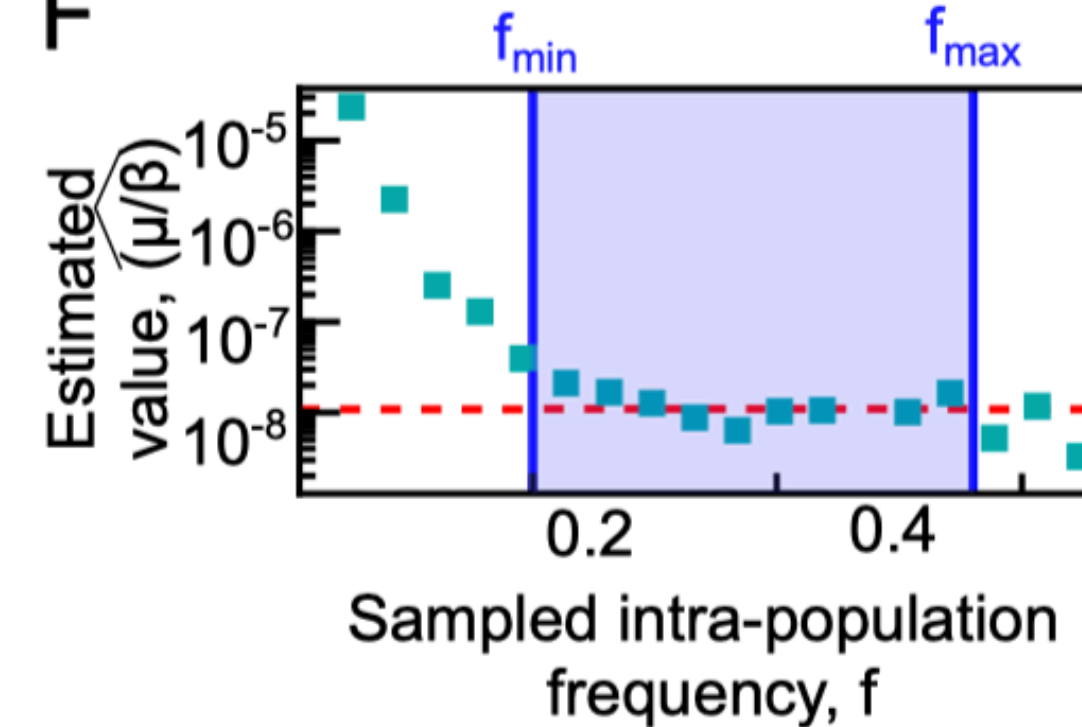
CRC0282PR (MSI)



E



F



G

